

- 4 Tetanus is a disease caused by infection with the bacterium *Clostridium tetani*.

A vaccine is available to provide protection against tetanus.

- (a) The passage outlines some information about tetanus and *C. tetani*.

Complete the passage by using the most appropriate scientific terms.

Tetanus is unlike other diseases for which vaccines are available because it cannot be passed from an infected person to another person. Tetanus is therefore **not** considered to be a disease. However, the bacterium *C. tetani* can be described as a because it causes a disease when it infects an individual. *C. tetani* can enter the human body when a person has a wound that makes contact with substances that contain *C. tetani*, such as soil or dust.

[2]

- (b) *C. tetani* releases a toxin, which produces the symptoms of tetanus in an infected person. The tetanus toxin is a protein, with part of its structure functioning as an enzyme.

Figure 4.1 shows the simplified structure of the tetanus toxin.

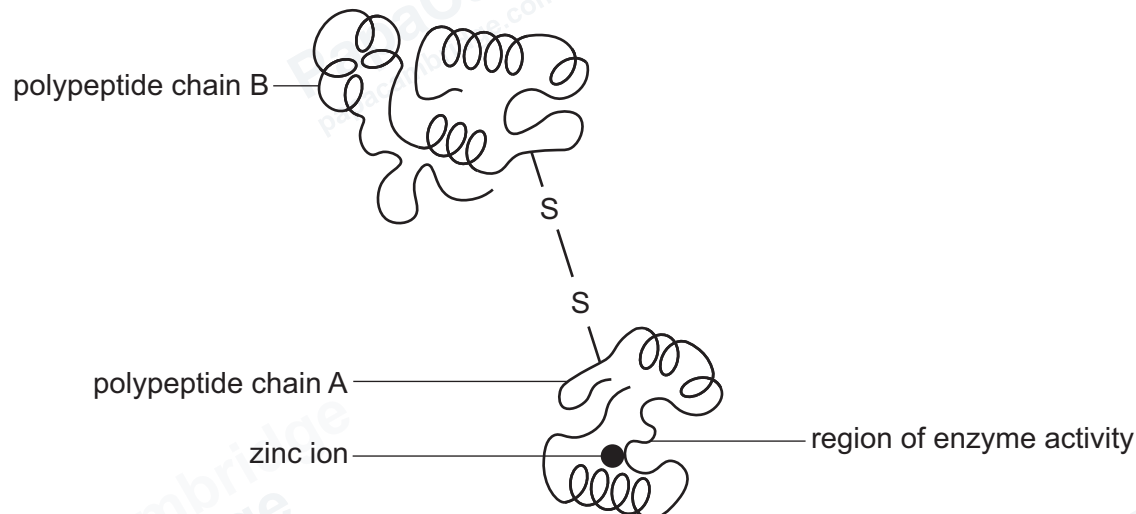


Figure 4.1



Describe the structure of the tetanus toxin shown in Figure 4.1.

In your answer, you should link the descriptions to the different levels of protein structure shown in Figure 4.1.

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..... [4]

- (c) The tetanus toxin enters nerve cells (neurones), where it causes the symptoms of the disease.

Figure 4.2 shows the entry of the tetanus toxin into a neurone.

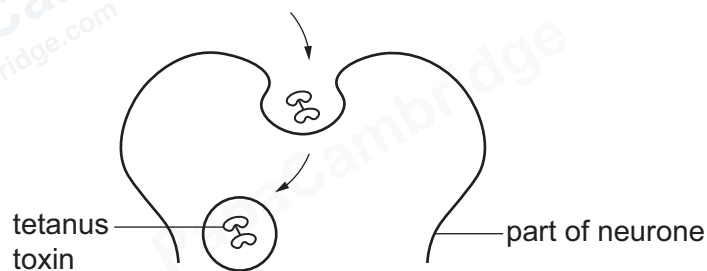


Figure 4.2

State the name of the process by which the toxin enters the neurone, as shown in Figure 4.2.

..... [1]

- (d) The symptoms of tetanus can develop in a person very quickly after infection by *C. tetani*.

People can gain immunity to infection by *C. tetani* in a number of different ways.

Table 4.1 describes three ways in which immunity to infection by *C. tetani* can develop in a person.

Complete Table 4.1 by naming the types of immunity that are described.

Table 4.1

description	types of immunity
Antibodies against tetanus toxin are passed from the mother to the fetus during pregnancy.	 and
Inactivated tetanus toxin is injected into a person.	 and
Anti-tetanus antibodies are injected into a person who has been exposed to <i>C. tetani</i> .	 and

[3]

- (e) Table 4.2 lists some events that occur during a primary immune response to infection by *C. tetani*.

Each event is represented by a letter.

Table 4.2

letter	event
A	antibodies are released into the circulation
B	antigens are presented on the surface of a macrophage
C	B-lymphocytes differentiate into plasma cells
D	T-helper cells produce cytokines that stimulate B-lymphocytes
E	T-helper cells are activated and divide by mitosis
F	pathogen enters macrophage by phagocytosis

State the sequence of letters that shows a correct order of events that occur during an immune response.

..... [3]

[Total: 13]



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